

NICHOLAS WILLIAMS

<https://nchlswllms.github.io> ♦ nw480@cam.ac.uk

PERSONAL INFORMATION

Full name: Nicholas James Williams
Date of birth: 11th July 1994
Place of birth: London, UK
Citizenship: UK

EDUCATION

Doctorate in Mathematics	<i>September 2018 – April 2022</i>
University of Cologne	<i>April 2021 – April 2022</i>
Started at University of Leicester	<i>September 2018 – April 2021</i>
Supervisor: Sibylle Schroll	
<i>Summa cum laude</i>	

Graduate Diploma in Quantitative Economics	<i>October 2017 – June 2018</i>
Birkbeck, University of London	
Distinction	

MMathPhil in Mathematics & Philosophy	<i>October 2012 – June 2016</i>
Worcester College, Oxford	
Double First Class Honours	

EMPLOYMENT

Research Associate	<i>October 2024 – September 2026</i>
University of Cambridge	
With Ailsa Keating	

Postdoctoral Research Associate	<i>September 2022 – September 2024</i>
Lancaster University	
With David Pauksztello	

JSPS Short Term Postdoctoral Research Fellow	<i>May 2022 – August 2022</i>
University of Tokyo	
With Osamu Iyama	

Graduate Teaching Assistant	<i>September 2021 – December 2021</i>
King's College London	

Guest Teacher	<i>September 2021 – December 2021</i>
London School of Economics	

Feedback Session Leader	<i>January 2019 – December 2020</i>
University of Leicester	

JOURNAL PUBLICATIONS

- | | |
|--|-------------|
| 10. A geometric perspective on the τ -cluster morphism category
With Sibylle Schroll, Aran Tattar, and Hipolito Treffinger
<i>Mathematische Zeitschrift</i> , Volume 312, article number 77 | <i>2026</i> |
| 9. The two higher Stasheff–Tamari orders are equal
<i>Journal of the European Mathematical Society</i> , Volume 28, no. 4, pp. 1391–1442 | <i>2026</i> |

8. Steenrod operations via higher Bruhat orders 2025
 With Guillaume Laplante-Anfossi
Proceedings of the London Mathematical Society, Volume 130, no. 2, e70024
7. A survey of congruences and quotients of partially ordered sets 2024
EMS Surveys in Mathematical Sciences, Volume 11, no. 1, pp. 153–203
6. Triangulations of prisms and preprojective algebras of type A 2024
 With Osamu Iyama
International Mathematics Research Notices, no. 13, 10236–10254
5. Stability spaces of string and band modules 2024
 With Sibylle Schroll, Aran Tattar, Hipolito Treffinger, and Yadira Valdivieso
Journal of Pure and Applied Algebra, Volume 228, no. 4, 107503
4. Quiver combinatorics and triangulations of cyclic polytopes 2023
Algebraic Combinatorics, Volume 6, no. 3, pp. 639–60
3. The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders 2023
Electronic Journal of Combinatorics, Volume 30, no. 1, P1.29
2. New interpretations of the higher Stasheff–Tamari orders 2022
Advances in Mathematics, Volume 407, 108552
1. The combinatorics of tensor products of higher Auslander algebras of type A 2021
 With Jordan McMahon
Glasgow Mathematical Journal, 63(3), pp. 526–546

PEER-REVIEWED CONFERENCE PROCEEDINGS

4. From higher Bruhat orders to Steenrod cup- i coproducts 2024
 With Guillaume Laplante-Anfossi
Proceedings of the 36th International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 91B, Art. 10
3. The higher Stasheff–Tamari orders in representation theory 2023
Representations of Algebras and Related Structures
2. Studying triangulations of even-dimensional cyclic polytopes via directed graphs 2023
Proceedings of the 35th International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 89B, Art. 4
1. The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders 2021
Proceedings of the 33rd International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 85B, Art. 6

OTHER CONFERENCE PROCEEDINGS

2. Donaldson–Thomas invariants for the Bridgeland–Smith correspondence 2024
Cluster Algebras and Its Applications
Oberwolfach Reports, No. 2/2024
1. Triangulations of cyclic polytopes and the higher Auslander algebras of type A 2022
Various issues on representation theory and related topics
RIMS Kōkyūroku, No. 2234

OTHER PUBLICATIONS

2. Microthesis: Triangulations of cyclic polytopes in representation theory 2022
London Mathematical Society Newsletter, No. 501, 33–34

- | | |
|--|------|
| 1. Higher-dimensional combinatorics in representation theory
Doctoral thesis, University of Cologne | 2022 |
|--|------|

PREPRINTS

- | | |
|---|------|
| 5. Relations between categorifications of higher-dimensional type A cluster combinatorics
With Mikhail Gorsky
arXiv:2605.27263 | 2026 |
| 4. Oriented matroids from type A cluster categories
arXiv:2509.07883 | 2025 |
| 3. Preorders on maximal chains:
hyperplane arrangements, Cambrian lattices, and maximal green sequences
With Mikhail Gorsky
arXiv:2506.08858 | 2025 |
| 2. Donaldson–Thomas invariants for the Bridgeland–Smith correspondence
With Omar Kidwai
arXiv:2401.10093 | 2024 |
| 1. A structural view of maximal green sequences
With Mikhail Gorsky
arXiv:2301.08681 | 2023 |

PRIZES AND SCHOLARSHIPS

- | | |
|--|------|
| Gibbs prize for Mathematics & Philosophy
University of Oxford
For top performance in Mathematics papers | 2016 |
| Gibbs prize for Mathematics & Philosophy
University of Oxford
For top performance in Mathematics papers | 2015 |
| Academic scholarship
Worcester College, Oxford | 2013 |

INVITED CONFERENCE TALKS (IN PERSON)

- | | |
|---|----------------|
| Higher Ginzburg DG algebras from dimer models
and their symplectic incarnations
<i>Symplectic topology meets representation theory</i>
University of Cambridge | April 2026 |
| Donaldson–Thomas invariants for the Bridgeland–Smith correspondence
<i>Algebraic Geometry Days in Glasgow</i>
University of Glasgow | May 2024 |
| Triangulations of cyclic polytopes through the lens of reconfiguration
<i>Minisymposium on Combinatorial Reconfiguration</i>
<i>International Congress on Industrial and Applied Mathematics</i>
Waseda University, Tokyo | August 2023 |
| A structural view of maximal green sequences
<i>Cluster structures in the North</i>
University of Central Lancashire | June 2023 |
| A structural approach to maximal green sequences
<i>Silting theory, algebras and representations</i>
Charles University, Prague | September 2022 |

INVITED CONFERENCE TALKS (VIRTUAL)

- Oriented matroids and maximal green sequences of type A cluster algebras** *January 2026*
Flash Talks in Representation Theory
- An algebraic interpretation of the higher Stasheff–Tamari orders** *November 2020*
International Conference on Representations of Algebras
Selected to give talk by scientific committee based on research snapshot

CONTRIBUTED CONFERENCE TALKS (IN PERSON)

- Oriented matroids from type A cluster categories** *July 2026*
International Conference on Representations of Algebras
Université Grenoble Alpes
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *August 2024*
International Conference on Representations of Algebras
Shanghai Jiao Tong University
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *February 2024*
Clusters and their Applications in the North
University of Glasgow
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *January 2024*
Cluster Algebras and Its Applications
Mathematisches Forschungsinstitut Oberwolfach
- Triangulations of prisms and preprojective algebras of type A** *July 2023*
Combinatorial representation theory in Leeds
University of Leeds
- A structural view of maximal green sequences** *April 2023*
Maurice Auslander Distinguished Lectures and International Conference
Woods Hole, Massachusetts
- Cyclic polytopes and representation theory** *December 2022*
Representation theory of algebras session
Canadian Mathematical Society Winter Meeting, Toronto
- Algebraic interpretation of the higher Stasheff–Tamari orders** *July 2022*
Various issues on representation theory and related topics
Research Institute for Mathematical Sciences, Kyoto University
- Higher analogues of Grassmannian clusters** *April 2019*
Maurice Auslander Distinguished Lectures and International Conference
Woods Hole, Massachusetts
- Higher analogues of Grassmannian clusters** *April 2019*
New Connections and Directions in Representation Theory
University of Leeds
- Higher analogues of Grassmannian clusters** *April 2019*
Tropical Geometry meets Representation Theory
University of Leicester

CONTRIBUTED CONFERENCE TALKS (VIRTUAL)

The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders

January 2022

Formal Power Series and Algebraic Combinatorics

Bar-Ilan University

SEMINAR TALKS (IN PERSON)

Higher Ginzburg DG algebras from dimer models and their symplectic incarnations

March 2026

MAGIC seminar

Imperial College London

Cluster algebras, cluster categories, and higher cluster categories

January 2026

Algebra and Representation Theory Seminar

University of Cambridge

Mutation des carquois et groupes de tresses

December 2025

Séminaire de géométrie, représentations, algèbre, analyse, logique

Université de Caen Normandie

Steenrod operations via higher Bruhat orders

December 2024

Algebra, Number Theory, Logic and Representation theory seminar

University of East Anglia

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

October 2024

Algebraic Geometry Seminar

University of Cambridge

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

June 2024

Geometric, Algebraic, and Physical structures around the moduli of Meromorphic

Quadratic Differentials

Simons Center for Geometry and Physics

Stony Brook University

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

April 2024

Leeds Algebra Seminar

University of Leeds

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

April 2024

Algebra and Number Theory Seminar

University of Vienna

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

March 2024

Paris Algebra Seminar

Institut de Mathématiques de Jussieu-Paris Rive Gauche

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

October 2023

Sheffield Geometry and Physics seminar

University of Sheffield

Donaldson–Thomas invariants for the Bridgeland–Smith correspondence

October 2023

EGRET seminar

University of Edinburgh

Donaldson–Thomas theory of quivers with potential

September 2023

Exact Asymptotics Theoretical Sciences Visiting Program

Okinawa Institute of Science and Technology

- Higher-dimensional cluster combinatorics and representation theory** *August 2023*
 OIST Representation Theory Seminar
 Okinawa Institute of Science and Technology
- Combinatorics of KP solitons** *August 2023*
 Exact Asymptotics Theoretical Sciences Visiting Program
 Okinawa Institute of Science and Technology
- A structural view of maximal green sequences** *February 2023*
 Algebra Group Seminar
 University of Cologne
- The higher Stasheff–Tamari orders on triangulations of cyclic polytopes** *November 2022*
 Séminaire du LaCIM
 Université du Québec à Montréal
- Cyclic polytopes and representation theory** *November 2022*
 Geometry and Mathematical Physics Seminar
 University of Birmingham
- Cyclic polytopes and representation theory** *October 2022*
 Pure Mathematics Seminar
 Lancaster University
- Mutation in higher cluster categories** *October 2022*
 Seminar@Aarhus
 Aarhus University
- Equivalence of maximal green sequences** *April 2022*
 Working Seminar in Algebra and Geometry
 University of Cologne
- Equivalence of maximal green sequences** *February 2022*
 Paris Algebra Seminar
 Institut de Mathématiques de Jussieu-Paris Rive Gauche
- An algebraic interpretation of the higher Stasheff–Tamari orders** *November 2019*
 Pure Mathematics Seminar
 University of East Anglia
- Higher analogues of Grassmannian clusters** *April 2019*
 Algebra and Number Theory Seminar
 University of Graz

SEMINAR TALKS (VIRTUAL)

- Oriented matroids and maximal green sequences
of type A cluster algebras** *April 2026*
 Seminario de Álgebra homológica e hierbas afines
 University of Buenos Aires
- Oriented matroids and maximal green sequences
of type A cluster algebras** *October 2025*
 Combinatorics and Representation Theory seminar
 Korea Institute for Advanced Study
- Steenrod operations via higher Bruhat orders** *May 2025*
 Longitudinal Algebra & Geometry Open Online Seminar (LAGOON)

Oriented matroids from cluster-tilting objects in type A cluster categories *March 2025*

New Directions in Group Theory & Triangulated Categories

Donaldson–Thomas invariants for the barbell quiver with potential

June 2024

Cologne algebra seminar

University of Cologne

Mutating cluster-tilting objects in $(d + 2)$ -angulated cluster categories

June 2022

FD seminar

An algebraic interpretation of the higher Stasheff–Tamari orders

January 2022

Aachen–Bochum–Cologne Darstellungstheorie Seminar

RWTH Aachen University, Ruhr-University Bochum, and University of Cologne

Cyclic polytopes and higher Auslander–Reiten theory

December 2021

Tokyo–Nagoya algebra seminar

University of Tokyo and Nagoya University

The two higher Stasheff–Tamari orders are equal

May 2021

Working seminar in algebra and geometry

University of Cologne

The higher Stasheff–Tamari orders in representation theory

April 2021

Paris algebra seminar

Institut de Mathématiques de Jussieu-Paris Rive Gauche

New interpretations of the higher Stasheff–Tamari orders

March 2021

Diskrete Mathematik / Geometrie Forschungsseminar

TU Berlin

LECTURE SERIES (IN PERSON)

**Introduction to Donaldson–Thomas theory
for quivers with potential**

October 2023 – February 2024

Lancaster University

12 lectures

LECTURE SERIES (VIRTUAL)

Cyclic polytopes and higher Auslander–Reiten theory

June – July 2022

Tokyo–Nagoya algebra seminar

University of Tokyo and Nagoya University

3 lectures

POSTERS

From higher Bruhat orders to Steenrod cup- i coproducts

July 2024

Formal Power Series and Algebraic Combinatorics

Ruhr-Universität Bochum

**Studying triangulations of even-dimensional cyclic polytopes
via directed graphs**

July 2023

Formal Power Series and Algebraic Combinatorics

University of California, Davis

An algebraic interpretation of the higher Stasheff–Tamari orders*June 2021**Young Researchers in Mathematics*

Online

Winner of 1st prize in poster competition**An algebraic interpretation of the higher Stasheff–Tamari orders***May 2021**Advances in Representation Theory of Algebras*

Online

New interpretations of the higher Stasheff–Tamari orders*January 2021**Combinatorial algebra meets algebraic combinatorics*

Université du Québec à Montréal

Online

An algebraic interpretation of the higher Stasheff–Tamari orders*November 2020**International Conference on Representations of Algebras*

“Research snapshot”

Online

New interpretations of the higher Stasheff–Tamari orders*September 2020**Virtual Heidelberg Laureate Forum*

Online

Representation theory and Grassmannians*March 2019**STEM for Britain*

Poster competition at UK Parliament

GRANTS AND AWARDS

London Mathematical Society Scheme 1

October 2023

Conference grants

Silting, simple-minded, and stability

Co-organised with Raquel Coelho Simões and David Pauksztello

GBP 4,000

Japan Society for the Promotion of Science

December 2021

Short Term Postdoctoral Research Fellowship

JPY 5,384,000

12 months awarded but 3 months taken

London Mathematical Society Scheme 8

April 2020

Postgraduate research conference grants

Categorifications in Representation Theory

Co-organised with Aran Tattar

GBP 2,500

ORGANISATION

*Simple-mindedness, silting, and stability**July 2024*

Co-organised with Raquel Coelho Simões and David Pauksztello

Ambleside Campus, University of Cumbria

*Köln Algebra and Representation Theory seminar**April 2021 – May 2022*

Co-organised with Daniel Schaefer

University of Cologne

Online

Categorifications in Representation Theory
Co-organised with Aran Tattar
Online

September 2020

Mathematics PhD seminar
University of Leicester

September 2019 – August 2020

*Interactions between Representation Theory
and Homological Mirror Symmetry*
Local organising committee
University of Leicester

May 2019

RESEARCH STAYS

Université de Caen Normandie
Visiting: Matthew Pressland
1 week

December 2025

Simons Center for Geometry and Physics
Stony Brook University
*Geometric, Algebraic, and Physical structures around the moduli of Meromorphic
Quadratic Differentials*
Invited participant
2 weeks

June 2024

University of Copenhagen
Visiting: Guillaume Laplante-Anfossi
Host: Nathalie Wahl
1 week

January 2024

Okinawa Institute of Science and Technology
Exact Asymptotics: From Fluid Dynamics to Quantum Geometry
Theoretical Sciences Visiting Program
Invited participant
2 months

August – October 2023

Université du Québec à Montréal
Visiting: Hugh Thomas
2 weeks

November 2022

University of Aarhus
Visiting: Raphael Bennett-Tennenhaus and Amit Shah
1 week

October 2022

University of Cologne
Visiting: Sibylle Schroll and Aran Tattar
1 week

April 2021

Isaac Newton Institute for Mathematical Sciences
University of Cambridge
Cluster algebras and representation theory
Invited participant
1 week

December 2021

University of Graz
Visiting: Jordan McMahon
Host: Karin Baur
1 week

April 2019

REFEREEING

Algebra and Number Theory, Algebraic Geometry and Physics, Annals of Representation Theory, Communications in Algebra, Compositio Mathematica, Discrete Mathematics (x2), Electronic Journal of Combinatorics (x2), Electronic Research Archive, Indagationes Mathematicae, Journal of Algebra (x2), Journal of Algebraic Combinatorics, Selecta Mathematica

JSPS Short Term Pre/Postdoctoral Fellowships (x9)

Quick opinions: Inventiones Mathematicae, Proceedings of the Edinburgh Mathematical Society, Selecta Mathematica

REVIEWING

AMS MathSciNet *October 2022 – present*
15 reviews written

zbMATH *September 2020 – present*
17 reviews written

TEACHING

Part II Number Fields *January 2026 – May 2026*
Supervisor
University of Cambridge

Part III Commutative Algebra *October 2025 – July 2026*
Lecturer for 24-lecture course taken by 62 students
Other duties: teaching problem sheet classes and setting examination
University of Cambridge

Part IB Groups, Rings, and Modules *February 2025*
Lecture cover: 1 lecture
University of Cambridge

Part II Algebraic Geometry *January 2025 – May 2025*
Supervisor
University of Cambridge

Real analysis *September 2021 – December 2021*
Class teacher (online)
King's College London

MA107: Quantitative methods *September 2021 – December 2021*
Class teacher
London School of Economics

Calculus & analysis *October 2020 – December 2020*
Class teacher (online)
University of Leicester

Second-year algebra *January 2020 – April 2020*
Class teacher
University of Leicester

Calculus & analysis *October 2019 – April 2020*
Class teacher
University of Leicester

Elements of number theory

Class teacher

University of Leicester

*October 2019 – December 2019***Calculus & analysis**

Class teacher

University of Leicester

*January 2019 – March 2019***STUDENT SUPERVISION**

Part III essays (Master's project)

2025 – 2026

University of Cambridge

Students: Ariff Bin-Johan and David Lumsden

Cambridge Summer Research in Mathematics

*2025*Project: *Combinatorics of KP solitons*

Students: Lilian Contius, Chris Yuzhen Yang

COMMITTEE ROLES

College Affiliation Officer

February 2026 – present

Postdocs of Cambridge Society

Mathematics & Philosophy Representative

2015 – 2016

Joint Consultative Committee with Undergraduates

University of Oxford

Undergraduate Representative

2015 – 2016

Joint Committee for Mathematics and Philosophy

University of Oxford

Undergraduate Representative

2015 – 2016

Mathematics Department Art Committee

University of Oxford

OUTREACH

Target Oxbridge lecture*April 2026*

One-hour lecture giving a flavour of what Maths at Cambridge is like, delivered to high-achieving Year 12 students from a Black heritage background

University of Leicester Representative

*November 2018, 2019, 2020**Research in the UK afternoon*

University of Cambridge

Student Helper

*March 2013, 2014, 2015**Oxford and Cambridge Student Conferences***MEMBERSHIPS**

Trinity College Postdoctoral Society*January 2025 – present***London Mathematical Society***June 2019 – present*

Associate member

3 reports on conferences written for the LMS newsletter

OTHER

Interviewer for undergraduate mathematics admissions

December 2025

Two full days of interviews

Emmanuel College, Cambridge

CATAM examiner

February – May 2025

Marker for undergraduate computer projects

University of Cambridge

Programming

Python, Sage

Git

L^AT_EX

Languages

(CEFR levels)

English	Native
French	B1
German	B1
Japanese	B1